This version covers the full, finished project: data cleaning, network trends, store segmentation, promo A/B test, holidays/seasonality, store ranking, and the interactive dashboard.

[0:00-0:40] Introduction

Good morning. My final project is about the Rossmann Store Sales dataset — over one million daily sales records from 1,115 drugstores in Germany, over about two and a half years.

I chose this dataset because I have real experience working in retail. My goal was not just to describe the data but also answer real business questions, and test every claim with statistics, not just charts. I tried to work like a real data analyst: clean the data, write clear hypotheses, and stay honest about what the data can and cannot prove.

[0:25-0:55] Data Preparation

Before analysis, I combined sales data with store information and checked data quality. I found one real problem: some stores had a opening year of 1910 — it is clearly a placeholder, not a real date. I marked it as missing instead of letting it damage later results. I also built new features, like sales per customer, always checking for divide-by-zero errors.

[0:55-1:35] Network-Level Trends

At the network level, I confirmed two patterns. December is the best month, about 20% higher than November, the second-best month. And sales change by day of week — Monday is highest, and this is statistically significant.

[SWITCH TO DASHBOARD → Overview tab] "Here's that same trend live — you can see the December peak right here." *(Point, then switch back.)*

[1:35-2:15] Store Segmentation

Store type "b" has the highest average sales — but only 17 of 1,115 stores are this type. So this is not a strong basis for a company-wide decision — it's a signal to study these stores individually. Stores with a full product assortment sell more per customer, and this is real and significant. But distance to the nearest competitor showed almost no connection to sales.

[SWITCH TO DASHBOARD → Segmentation tab] "Here's that store-type-b warning, right in the dashboard." *(Point, then switch back.)*

[2:15-3:05] Promotion A/B Test

The center of my project is a proper A/B test on promotions. Honest point first: stores choose when to run a promotion, so I can't say for certain that promotions cause higher sales — only that they happen together. With that said, the result is strong: sales are 39% higher on promo days, and this is a medium-to-large effect, not just a statistically significant one. About 20% comes from more customers, and 14% from bigger baskets — so promotions work two ways at once. The effect size also differs by store type, so one single promo strategy does not fit every store. One surprising result: stores in the long-term "Promo2" campaign actually show lower sales — most likely because weaker stores are the ones that join these campaigns in the first place, not because the campaign hurts them.

[SWITCH TO DASHBOARD → Promotions tab] "Here's that promo effect by store type, and the Promo2 surprise below it." *(Point, then switch back.)*

[3:05-3:40] Holidays and Seasonality

Pre-holiday days show a real sales peak — about 12% higher than normal days, confirming the "pre-Christmas rush" idea. But I found something I didn't expect: on the holiday day itself, the few stores that stay open actually show higher sales, not lower. This makes sense once you think about it — almost every store is closed, so the handful that stay open, likely stations or tourist spots, absorb all the demand from people who still need to shop. School holidays also lift sales, but the effect size is different for each store type.

[SWITCH TO DASHBOARD → Holidays & Seasonality tab] "You can see that holiday-day surprise explained right here." *(Point, then switch back.)*

[3:40-4:15] Store Ranking

I ranked all 1,115 stores by sales, stability, and year-over-year growth. The most important finding: a store's total sales and its growth rate are almost unrelated — the top 20 stores by sales and the top 20 by growth had zero overlap. Big and successful today does not mean fast-growing. I also found the whole network's sales actually dropped 5.3% from 2013 to 2014, then recovered 6.6% into 2015 — so any single store's performance has to be mentioned against that network-wide trend, not in isolation.

[SWITCH TO DASHBOARD → Store Explorer tab] "And this is the full ranking — every store, searchable and sortable." *(Point for 3-4 seconds, then switch back.)*

[4:15-5:00] Closing

To summarize: this project shows the simple story is often not the full story. The "best" store type is really just 17 stores. Promotions really work, but not equally everywhere. A holiday campaign that looks harmful is probably a symptom of a weak store, not a reason. And the biggest stores aren't the fastest-growing ones. All of this came from testing ideas properly with statistics — Mann-Whitney tests, Cohen's d, confidence intervals — not from just looking at a chart and guessing.

In this project I wanted to build a tool that uses approaches we learned in this course: cleaning, testing hypotheses, and visualizing results. And this tool is not only for Rossmann. The same approach — the notebooks, the statistical tests, the dashboard — can be reused for any retail chain, just by pointing it at a different sales dataset.

That is really the main takeaway of this whole project. Thank you — I'm happy to answer questions.